

OG-Core Examples: TFP, Productivity, Mortality Shocks

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Examples

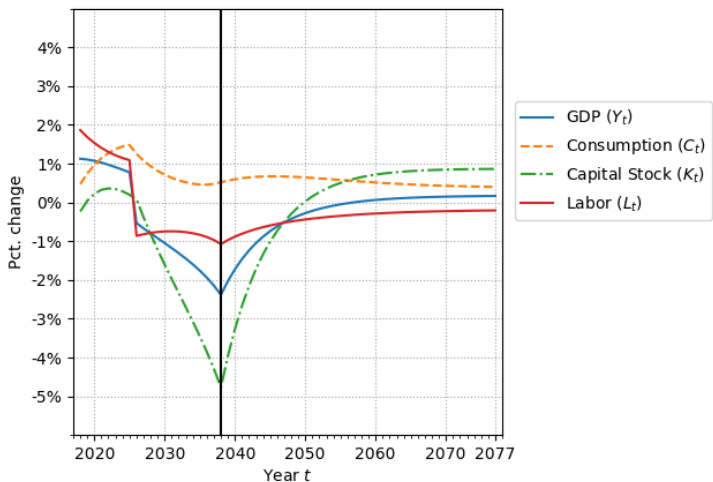
- Tax Cuts and Jobs Act (TCJA) Dynamic Analysis
- Revenue and Macroeconomic Effects of a 70% Marginal Tax Rate
- A Detailed Macroeconomic Analysis of President Biden's 2020 Campaign Tax Proposals
- The Macroeconomics of Improving Biological Aging

TCJA Dynamic Analysis

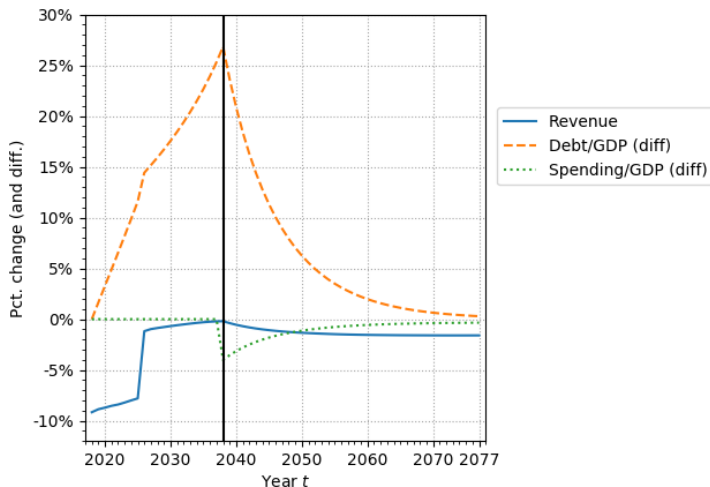
DeBacker, Jason and Richard W. Evans, “[Dynamic Analysis of Tax Cuts and Jobs Act](#),” *Quantitative Notes*, 2018-1, (February 2, 2018).

- GDP growth in first 8 years (1-2% increase)
- Significant crowding out as debt increases

TCJA Macro variables, closed economy



TCJA Fiscal variables, closed economy



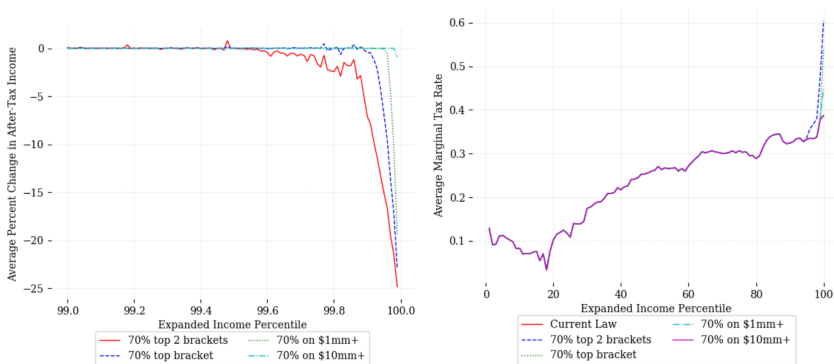
TCJA Dynamic Analysis

DeBacker, Jason and Anderson Frailey, “[Revenue and Macroeconomic Effects of a 70% Marginal Tax Rate](#),” *Quantitative Notes*, 2019-1, (March 4, 2019).

- Can raise between \$5B and \$250B per year
- Reduce GDP in short run between 0.1% and 1.7%

70 Percent Incidence

Figure 2. Distributional Impacts of a 70% Top Rate

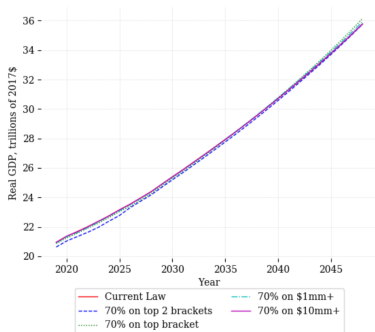


(a) Percentage Change in After-tax Income

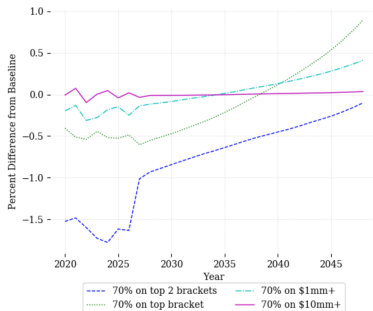
(b) Average Marginal Tax Rates

70 Percent GDP change

Figure 3. Real GDP, 2019-2049



(a) Gross Domestic Product



(b) Percentage Differences from Baseline GDP

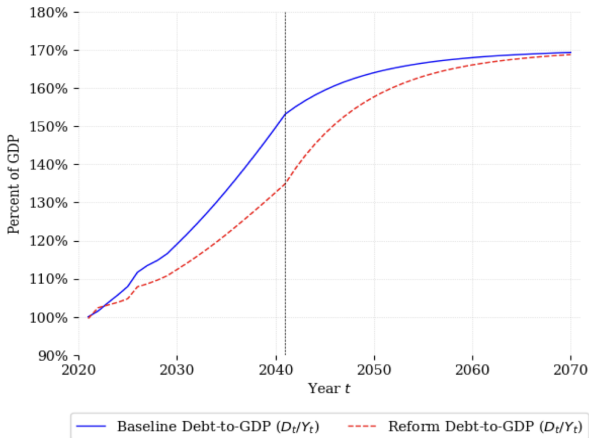
Analysis of President Biden's Campaign Proposals

DeBacker, Jason, Richard Evans, and Benjamin Page, “[A Detailed Macroeconomic Analysis of President Biden's 2020 Campaign Tax Proposals](#),” *Tax Policy Center Working Paper*, July, 2021.

- Debt reduction crowds in private capital in medium term
- We provide a sensitivity analysis to various parameter values

Biden Dynamic Analysis

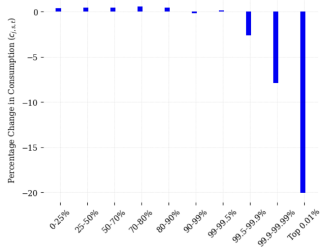
Figure 6: Debt-to-GDP Ratio, Current Law Baseline vs. Biden Proposals



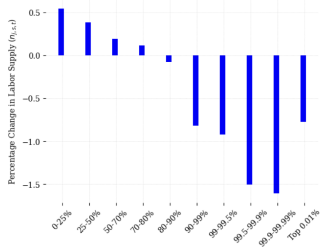
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Biden Dynamic Analysis

Figure 8: Behavioral Responses by Lifetime Income Group, 2021-2031



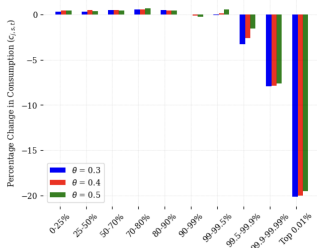
(a) Consumption



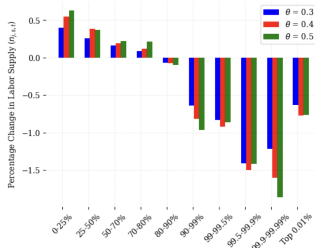
(b) Labor Supply

Biden Dynamic Analysis

Figure 1: Behavioral Responses by Lifetime Income Group, 2021-2031, Frisch Elasticity



(a) Consumption



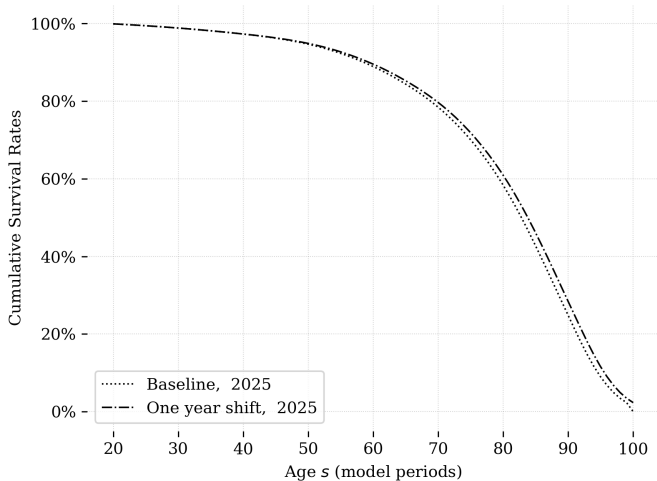
(b) Labor Supply

The Macroeconomics of Improving Biological Aging

DeBacker, Jason, Richard Evans, and Raiany Romanni, “The Macroeconomics of Improving Biological Aging”, *Working Paper*, July, 2024.

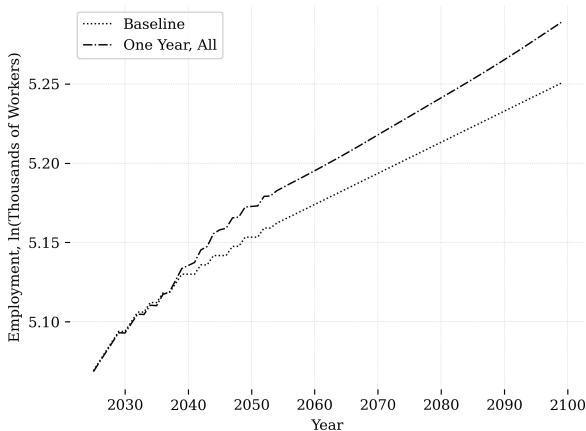
- Improving health outcomes leads to large macroeconomic benefits
- The time horizons over which health impacts affect the economy can be many decades

Changing Mortality



Employment Effects

Employment Paths, One Year Shift in Biological Age



GDP Effects

NPV of Change in GDP in Trillions of Constant 2025\$, One Year Shift in Biological Age

Discount Rate	Mortality	Productivity	Fertility	All
2%	98.52	35.75	73.21	209.86
4%	26.47	10.32	13.79	51.20
6%	8.63	3.69	2.77	15.29
Model r	10.40	4.47	3.25	18.06